Fluid Antenna-Enabled Physical Layer Security for UAV Communications

(Invited Paper)

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***Abstract*—This paper investigates a fluid antenna (FA)-enabled secure unmanned aerial vehicle (UAV) communication system to enhance physical-layer security (PLS) against eavesdropping with location uncertainty. We formulate a secrecy sum-rate maximization problem by jointly optimizing the transmit beamforming and the FA positions. To solve this non-convex problem with highly coupled variables, an alternating optimization (AO)-based algorithm is proposed. Simulation results show that the proposed scheme significantly outperforms benchmarks, demonstrating the advantage of FA-enabled spatial flexibility in secure UAV communications.**

***Index Terms*—Fluid antenna, antenna movement delay, transmit beamforming, secure communications.**

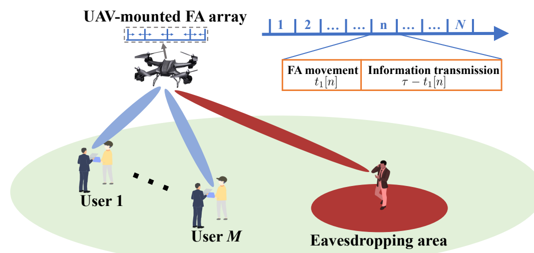

Fig. 1: Illustration of the considered FA-enabled secure UAV communication system with multiple users.

I. Introduction

Unmanned aerial vehicles (UAVs) have emerged as important enablers of next-generation wireless communication systems due to their high mobility and flexible deployment capabilities [1]. As key components of low-altitude wireless networks (LAWNs), UAVs can provide on-demand air-to-ground (A2G) communication services in remote or infrastructure-deficient areas and serve as temporary aerial base stations (BSs) to relieve traffic congestion in hotspot regions [2]–[4]. Benefiting from line-of-sight (LoS)-dominant channels, UAV-enabled communications can significantly enhance spectral efficiency and reduce transmission latency [5].

Despite these merits, the broadcast nature and LoS dominance of A2G channels render UAV communications highly susceptible to eavesdropping [6]. This vulnerability necessitates physical-layer security (PLS) techniques, which bolster secrecy rates via wireless channel randomness [7]. Unlike traditional cryptography, PLS offers a lightweight, adaptive solution tailored for dynamic UAV environments [8], [9]. Nevertheless, conventional fixed-position antennas (FPAs) suffer from constrained spatial adaptability, failing to fully harvest the spatial diversity inherent in dynamic scenarios [10], [11]. As a transformative alternative, fluid antennas (FAs) allow antenna elements to dynamically reposition within a continuous region [12]–[14]. By reconfiguring antenna geometry, FA-enabled systems can sharpen beamforming, reinforce legitimate links, and mitigate information leakage, unlocking new paradigms for secure UAV communications [15], [16].

Motivated by this, this paper investigates an FA-enabled secure UAV communication system, where a UAV with a FA array serves multiple ground users in the presence of eavesdroppers. To enhance overall secrecy performance, a secrecy sum-rate maximization problem is formulated by jointly optimizing the transmit beamforming and the FA positions. Moreover, a robust design considers multiple candidate eavesdropper locations within the uncertainty region to ensure secure communication in the worst-case scenario. The resulting non-convex optimization problem is efficiently solved via an alternating optimization (AO)-based algorithm, which yields a high-quality suboptimal solution. Simulation results demonstrate that the proposed scheme significantly outperforms conventional benchmark methods.

II. System Model and Problem Formulation

A. Network Model

As illustrated in Fig. 1, we consider an FA-enabled secure UAV communication system, where a UAV equipped with an FA array with K elements simultaneously serves M single-antenna users in the presence of a single-antenna eavesdropper. The UAV is assumed to know the locations of all users, while only the eavesdropping region $\mathcal{E}$ containing the eavesdropper is available. Consider a mission period $[0, T]$ divided into N slots ($\tau = T/N$). The UAV flies at a constant altitude H with time-varying horizontal position $\mathbf{q}[n]$. The ground positions of

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users and the eavesdropper are denoted as $\mathbf{q}_m[n]$ and $\mathbf{q}_{\mathrm{E}}[n]$. Similar to [15], the channel vectors from the UAV to user m and to the eavesdropper during time slot n are denoted as $\mathbf{h}_m[n]$ and $\mathbf{h}_{\mathrm{E}}[n]$, respectively, i.e., $\mathbf{h}_{\tilde{m}}[n] = \sqrt{\beta_{\tilde{m}}[n]}\mathbf{a}_{\tilde{m}}[n]$, where $\tilde{m} \in \mathcal{M}\bigcup\{\mathrm{E}\}$. $\beta_{\tilde{m}}[n] = \frac{h_0}{d_{\tilde{m}}^2[n]}$ denotes the large-scale path loss, where h_0 denotes the channel power gain and $d_{\tilde{m}}[n] = \sqrt{\|\mathbf{q}[n]-\mathbf{q}_{\tilde{m}}[n]\|^2 + H^2}$ denotes the distance. $\mathbf{a}_{\tilde{m}}[n] = \left[\mathrm{e}^{\mathrm{j}\frac{2\pi}{\lambda}x_1[n]\sin(\theta_{\tilde{m}}[n])}, \ldots, \mathrm{e}^{\mathrm{j}\frac{2\pi}{\lambda}x_K[n]\sin(\theta_{\tilde{m}}[n])}\right]^{\mathsf{T}}$ denotes the steering vector, where $x_k[n]$ is the FA k position and $\theta_{\tilde{m}}[n] = \arcsin\left(\frac{H}{\sqrt{\|\mathbf{q}[n]-\mathbf{q}_{\tilde{m}}[n]\|^2+H^2}}\right)$ denotes the vertical angle of departure (AoD).

Let $s_m[n] \sim \mathcal{CN}(0,1)$ and $\mathbf{w}_m[n]$ denote the information signal for the user m and the corresponding transmit beamforming vector, respectively. The transmitted signal from the UAV is denoted as $\mathbf{x}_t[n] = \sum_{m=1}^{M}\mathbf{w}_m[n]s_m[n]$. Therefore, the received signal at the user/eavesdropper $\tilde{m}$ can be expressed as $y_{\tilde{m}}[n] = \sum_{i=1}^{M}\mathbf{h}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i[n]s_i[n] + \nu_{\tilde{m}}[n]$, where $\nu_{\tilde{m}}[n] \sim \mathcal{CN}(0,\sigma_{\tilde{m}}^2)$ is the additive white Gaussian noise (AWGN). Then, the resulting signal-to-interference-plus-noise ratio (SINR) of the user m is given by

$$\gamma_m[n] = \frac{\left|\mathbf{h}_m^{\mathsf{H}}[n]\mathbf{w}_m[n]\right|^2}{\sum\limits_{i\neq m}^{\mathcal{M}}\left|\mathbf{h}_m^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 + \sigma_m^2}, \tag{1}$$

and the resulting SINR at the eavesdropper for eavesdropping the signal intended for user m is given by

$$\gamma_{m,\mathrm{E}}[n] = \frac{\left|\mathbf{h}_{\mathrm{E}}^{\mathsf{H}}[n]\mathbf{w}_m[n]\right|^2}{\sum\limits_{i\neq m}^{\mathcal{M}}\left|\mathbf{h}_{\mathrm{E}}^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 + \sigma_{\mathrm{E}}^2}. \tag{2}$$

B. Problem Formulation

As illustrated in Fig. 1, we divide each time slot into two phases [17]. Therefore, the rate of the user m, the rate of the eavesdropper for eavesdropping the signal intended for the user m, and the corresponding secrecy rate of the user m are respectively given by

$$R_m[n] = (\tau - t_1[n])\log_2(1+\gamma_m[n]), \tag{3}$$
$$R_{m,\mathrm{E}}[n] = (\tau - t_1[n])\log_2(1+\gamma_{m,\mathrm{E}}[n]), \tag{4}$$
$$R_m^{\mathrm{sec}}[n] = \left[R_m[n] - R_{m,\mathrm{E}}[n]\right]^+, \tag{5}$$

where $[x]^+ = \max\{x, 0\}$.

In this paper, we aim to maximize the secrecy sum-rate of users by jointly optimizing the transmit beamforming $\{\mathbf{w}_m[n]\}$, and the FA positions $\{x_k[n]\}$. Mathematically, we can formulate the optimization problem as

$$(\mathbf{P1}): \max_{\mathbf{w}_m[n], x_k[n]} \sum_{m=1}^{M}\sum_{n=1}^{N} R_m^{\mathrm{sec}}[n] \tag{6a}$$
$$\text{s.t.} \sum_{m=1}^{M}\|\mathbf{w}_m[n]\|^2 \leq P_{\max},\ \forall n, \tag{6b}$$
$$0 \leq x_1[n] \leq \cdots \leq x_K[n] \leq D_{\mathrm{FA}},\ \forall n, \tag{6c}$$
$$x_k[n] - x_{k-1}[n] \geq d_{\min},\ \forall k \in \mathcal{K}\backslash\{1\}, \forall n, \tag{6d}$$
$$\frac{|x_k[n] - x_k[n-1]|}{\tilde{v}} \leq t_1[n],\ \forall n \in \mathcal{N}\backslash\{1\}, \forall k, \tag{6e}$$

where $\tilde{v}$ in (6e) denotes the FA movement speed. Constraint (6b) ensures that the total transmit power does not exceed the maximum power budget $P_{\max}$. Constraint (6c) ensures that the FA positions are ordered and confined within the moving region $[0, D_{\mathrm{FA}}]$. Constraint (6d) imposes a minimum separation distance $d_{\min}$ between adjacent FAs to prevent mutual coupling. Finally, constraint (6e) ensures that the movement delay of the FA k does not exceed $t_1[n]$.

III. AO Algorithm For Solving Problem (**P1**)

The UAV only knows the eavesdropping region $\mathcal{E}$, not its exact location. To ensure security under the worst-case scenario, J candidate locations are sampled within this region, and the one yielding the highest eavesdropping rate is treated as the effective eavesdropper location. Then, we have

$$-\sum_{n=1}^{N} R_{m,j}[n] \geq \eta_m,\ \forall j, \forall m, \tag{7}$$

where η_m denotes the introduced auxiliary variable. Besides, we introduce auxiliary variables $\omega_m[n]$ and $\varpi_m[n]$, satisfying

$$\sum_{i=1}^{\mathcal{M}}\left|\mathbf{h}_m^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 + \sigma_m^2 \geq \omega_m[n]\varpi_m[n], \tag{8}$$

$$\varpi_m[n] \geq \sum_{i\neq m}\left|\mathbf{h}_m^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 + \sigma_m^2. \tag{9}$$

Therefore, the lower bound for the term $R_m[n]$ is given by

$$\tilde{R}_m^{\mathrm{lb}}[n] = (\tau - t_1[n])\log_2(\omega_m[n]). \tag{10}$$

According to [18], we have

$$\omega_m[n]\varpi_m[n] \leq \frac{1}{2}\left(\frac{\omega_m^{(l)}[n]}{\varpi_m^{(l)}[n]}\varpi_m^2[n] + \frac{\varpi_m^{(l)}[n]}{\omega_m^{(l)}[n]}\omega_m^2[n]\right), \tag{11}$$

where $\omega_m^{(l)}[n]$ and $\varpi_m^{(l)}[n]$ are the local points in the l-th iteration. Therefore, constraint (8) can be reformulated as

$$\sum_{i=1}^{\mathcal{M}}\left|\mathbf{h}_m^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 + \sigma_m^2 \geq \frac{1}{2}\left(\frac{\omega_m^{(l)}[n]}{\varpi_m^{(l)}[n]}\varpi_m^2[n] + \frac{\varpi_m^{(l)}[n]}{\omega_m^{(l)}[n]}\omega_m^2[n]\right). \tag{12}$$

Similarly, we can derive the lower bound for $\{-R_{m,j}[n]\}$, denoted as

$$R_{m,j}^{\mathrm{lb}}[n] = (\tau - t_1[n])\log_2(\tilde{\omega}_{m,j}[n]), \tag{13}$$

along with the associated constraints

$$\sum_{i\neq m}^{\mathcal{M}}\left|\mathbf{h}_j^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 + \sigma_j^2 \geq \frac{1}{2}\left(\frac{\tilde{\omega}_{m,j}^{(l)}[n]}{\tilde{\varpi}_j^{(l)}[n]}\tilde{\varpi}_j^2[n] + \frac{\tilde{\varpi}_j^{(l)}[n]}{\tilde{\omega}_{m,j}^{(l)}[n]}\tilde{\omega}_{m,j}^2[n]\right), \tag{14}$$

$$\tilde{\varpi}_j[n] \geq \sum_{i=1}^{\mathcal{M}}\left|\mathbf{h}_j^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 + \sigma_j^2, \tag{15}$$

Then, constraint (7) can be reformulated as

$$\sum_{n=1}^{N}\tilde{R}_{m,j}^{\mathrm{lb}}[n] \geq \eta_m\,\forall j, \forall m. \tag{16}$$

In the end, the problem (**P1**) can be reformulated as

$$(\mathbf{P2}): \max_{\mathbf{w}_m[n], x_k[n], \Phi} \sum_{m=1}^{M} \left(\sum_{n=1}^{N} \tilde{R}_m^{\text{lb}}[n] + \eta_m \right) \tag{17a}$$
$$\text{s.t.} (6b) - (6e), (9), (12), (14) - (16),$$

where $\Phi \triangleq \{\eta_m, \omega_m[n], \varpi_m[n], \tilde{\omega}_{m,j}[n], \tilde{\varpi}_j[n]\}$. In the following subsection, we alternately optimize the transmit beamforming $\{\mathbf{w}_m[n]\}$, and the FA positions $\{x_k[n]\}$.

A. Transmit Beamforming Optimization

Given $\{x_k[n]\}$, the subproblem of (**P2**) for optimizing the transmit beamforming $\{\mathbf{w}_m[n]\}$ can be expressed as

$$(\mathbf{P3}): \max_{\mathbf{w}_m[n], \Phi} \sum_{m=1}^{M} \left(\sum_{n=1}^{N} \tilde{R}_m^{\text{lb}}[n] + \eta_m \right) \tag{18a}$$
$$\text{s.t.} (6b), (9), (12), (14) - (16).$$

Note that problem (**P3**) is still non-convex due to the non-convex constraints (12) and (14). Then, we apply the first-order Taylor expansion to obtain the lower bound for $\left|\mathbf{h}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2$, i.e.,

$$\begin{aligned} \left|\mathbf{h}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 \geq & \left|\mathbf{h}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i^{(l)}[n]\right|^2 + 2\Re\Big\{ \left(\mathbf{w}_i^{(l)}[n]\right)^{\mathsf{H}} \\ & \times \mathbf{h}_{\tilde{m}}[n]\mathbf{h}_{\tilde{m}}^{\mathsf{H}}[n] \left(\mathbf{w}_i[n] - \mathbf{w}_i^{(l)}[n]\right) \Big\} \\ & \triangleq C_{\tilde{m},i}^{(l)}[n], \end{aligned} \tag{19}$$

where $\tilde{m} \in \mathcal{M} \bigcup \mathcal{J}$. Therefore, constraints (12) and (14) can be reformulated as

$$\sum_{i=1}^{\mathcal{M}} C_{m,i}^{(l)}[n] + \sigma_m^2 \geq \frac{1}{2} \left(\frac{\omega_m^{(l)}[n]}{\varpi_m^{(l)}[n]} \varpi_m^2[n] + \frac{\varpi_m^{(l)}[n]}{\omega_m^{(l)}[n]} \omega_m^2[n] \right), \tag{20}$$

$$\sum_{i\neq m}^{\mathcal{M}} C_{j,i}^{(l)}[n] + \sigma_j^2 \geq \frac{1}{2} \left(\frac{\tilde{\omega}_{m,j}^{(l)}[n]}{\tilde{\varpi}_j^{(l)}[n]} \tilde{\varpi}_j^2[n] + \frac{\tilde{\varpi}_j^{(l)}[n]}{\tilde{\omega}_{m,j}^{(l)}[n]} \tilde{\omega}_{m,j}^2[n] \right). \tag{21}$$

With the above transformations, problem (**P3**) can be reformulated as the following convex optimization problem

$$(\mathbf{P3\text{-}1}): \max_{\mathbf{w}_m[n], \Phi} \sum_{m=1}^{M} \left(\sum_{n=1}^{N} \tilde{R}_m^{\text{lb}}[n] + \eta_m \right) \tag{22a}$$
$$\text{s.t.} (6b), (9), (15) - (16), (20) - (21).$$

B. FA Positions Optimization

Given $\{\mathbf{w}_m[n]\}$, the subproblem of (**P2**) for optimizing the FA positions $\{x_k[n]\}$ can be expressed as

$$(\mathbf{P4}): \max_{x_k[n], \Phi} \sum_{m=1}^{M} \left(\sum_{n=1}^{N} \tilde{R}_m^{\text{lb}}[n] + \eta_m \right) \tag{23a}$$
$$\text{s.t.} (6c) - (6e), (9), (12), (14) - (16).$$

Note that problem (**P4**) is non-convex due to the non-convex constraints (6e), (9), (12), (14), and (15). Firstly, we reformulate the constraint (6e) as

$$x_k[n-1] - t_1[n]\tilde{v} \leq x_k[n] \leq x_k[n-1] + t_1[n]\tilde{v}. \tag{24}$$

Next, we analyze the upper and lower bounds of the term $\left|\mathbf{a}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2, \tilde{m} \in \mathcal{M} \bigcup \mathcal{J}$ with respect to the FA positions $x_k[n]$. Let $w_{i,k}[n] = |w_{i,k}[n]|e^{j\angle w_{i,k}[n]}$ denote the k-th element of the beamforming vector $\mathbf{w}_i[n]$ in time slot n, with amplitude $|w_{i,k}[n]|$ and phase $\angle w_{i,k}[n]$. Moreover, we define $\vartheta_{\tilde{m}}[n] = \frac{2\pi}{\lambda}\sin(\theta_{\tilde{m}}[n])$. Then, by expanding the term $\left|\mathbf{a}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2$, we can obtain

$$\left|\mathbf{a}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 = \sum_{p=1}^{K}\sum_{q=1}^{K} |w_{i,p}[n]w_{i,q}[n]| \cos \Lambda_{\tilde{m},i,p,q}[n], \tag{25}$$

where $\Lambda_{\tilde{m},i,p,q}[n] = \vartheta_{\tilde{m}}[n](x_p[n] - x_q[n]) - (\angle w_{i,p}[n] - \angle w_{i,q}[n])$.

Due to $\left|\mathbf{a}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2$ is non-concave and non-convex with respect to $x_k[n]$, we construct a lower bound surrogate function to locally approximate it using its second-order Taylor expansion. Specifically, for a given $a \in \mathbb{R}$, we can obtain

$$\cos(a) \geq \cos(a_0) - \sin(a_0)(a-a_0) - \frac{1}{2}(a-a_0)^2 = f(a \mid a_0). \tag{26}$$

For given the local point $x_k^{(l)}[n]$, by letting $a_0 = \Lambda_{\tilde{m},i,p,q}^{(l)}[n]$ and $a = \Lambda_{\tilde{m},i,p,q}[n]$, we can obtain

$$\left|\mathbf{a}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 \geq \frac{1}{2}\mathbf{x}^{\mathsf{T}}[n]\mathbf{S}_{\tilde{m},i}^{\text{lb}}[n]\mathbf{x}[n] + \left(\mathbf{t}_{\tilde{m},i}^{\text{lb}}[n]\right)^{\mathsf{T}}\mathbf{x}[n] + u_{\tilde{m},i}^{\text{lb}}[n], \tag{27}$$

where $\mathbf{x}[n] = [x_1[n], \ldots, x_K[n]]^{\mathsf{T}}$. $\mathbf{S}_{\tilde{m},i}^{\text{lb}}[n]$, $\mathbf{t}_{\tilde{m},i}^{\text{lb}}[n]$, and $u_{\tilde{m},i}^{\text{lb}}[n]$ are defined as

$$\mathbf{S}_{\tilde{m},i}^{\text{lb}}[n] = -2\vartheta_{\tilde{m}}^2[n]\left(\mu_i[n]\mathsf{diag}\left(\tilde{\mathbf{w}}_i[n]\right) - \tilde{\mathbf{w}}_i[n]\tilde{\mathbf{w}}_i^{\mathsf{T}}[n]\right), \tag{28}$$

$$\begin{aligned} [\mathbf{t}_{\tilde{m},i}^{\text{lb}}]_p[n] =& 2\vartheta_{\tilde{m}}^2[n]\sum_{q=1}^{K}|w_{i,p}[n]w_{i,q}[n]|\left(x_p^{(l)}[n] - x_q^{(l)}[n]\right) \\ & - 2\vartheta_{\tilde{m}}[n]\sum_{q=1}^{K}|w_{i,p}[n]w_{i,q}[n]|\sin\left(\Lambda_{\tilde{m},i,p,q}^{(l)}[n]\right), \end{aligned} \tag{29}$$

$$\begin{aligned} u_{\tilde{m},i}^{\text{lb}}[n] =& \sum_{p=1}^{K}\sum_{q=1}^{K}|w_{i,p}[n]w_{i,q}[n]|\cos\left(\Lambda_{\tilde{m},i,p,q}^{(l)}[n]\right) + \vartheta_{\tilde{m}}[n] \\ & \times \sum_{p=1}^{K}\sum_{q=1}^{K}|w_{i,p}[n]w_{i,q}[n]|\sin\left(\Lambda_{\tilde{m},i,p,q}^{(l)}[n]\right)\left(x_p^{(l)}[n] - x_q^{(l)}[n]\right) \\ & - \frac{1}{2}\vartheta_{\tilde{m}}^2[n]\sum_{p=1}^{K}\sum_{q=1}^{K}|w_{i,p}[n]w_{i,q}[n]|\left(x_p^{(l)}[n] - x_q^{(l)}[n]\right)^2, \end{aligned} \tag{30}$$

where $\tilde{\mathbf{w}}_i[n] = [|w_{i,1}[n]|, |w_{i,2}[n]|, \cdots, |w_{i,K}[n]|]^{\mathsf{T}}$, and $\mu_i[n] = \sum_{p=1}^{K}|w_{i,p}[n]|$. By introducing the slack variable $\phi_{\tilde{m},i}^{\text{lb}}[n]$, we can obtain

$$\frac{1}{2}\mathbf{x}^{\mathsf{T}}[n]\mathbf{S}_{\tilde{m},i}^{\text{lb}}[n]\mathbf{x}[n] + \left(\mathbf{t}_{\tilde{m},i}^{\text{lb}}[n]\right)^{\mathsf{T}}\mathbf{x}[n] + u_{\tilde{m},i}^{\text{lb}}[n] \geq \phi_{\tilde{m},i}^{\text{lb}}[n]. \tag{31}$$

Since $\mathbf{S}_{\tilde{m},i}^{\text{lb}}[n]$ is negative semi-definite, constraint (31) is convex with respect to $\mathbf{x}[n]$. The complete proof is provided in [19].

Similarly, we can obtain

$$\begin{aligned} \left|\mathbf{a}_{\tilde{m}}^{\mathsf{H}}[n]\mathbf{w}_i[n]\right|^2 \leq & \frac{1}{2}\mathbf{x}^{\mathsf{T}}[n]\mathbf{S}_{\tilde{m},i}^{\text{ub}}[n]\mathbf{x}[n] + \left(\mathbf{t}_{\tilde{m},i}^{\text{ub}}[n]\right)^{\mathsf{T}}\mathbf{x}[n] + u_{\tilde{m},i}^{\text{ub}}[n], \\ & \leq \phi_{\tilde{m},i}^{\text{ub}}[n]. \end{aligned} \tag{32}$$

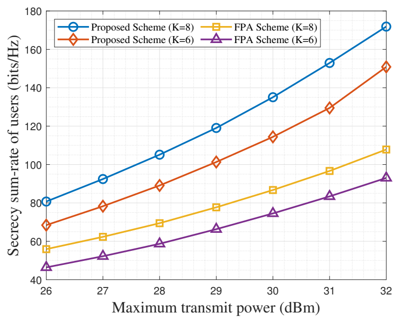

Fig. 2: The secure sum-rate versus the maximum transmit power.

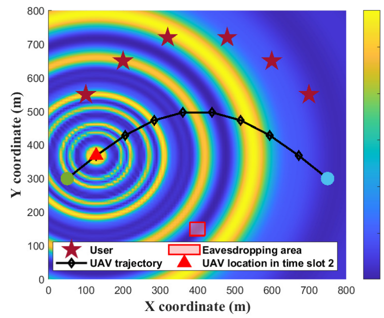

Fig. 3: Beamforming pattern gain achieved under the proposed scheme.

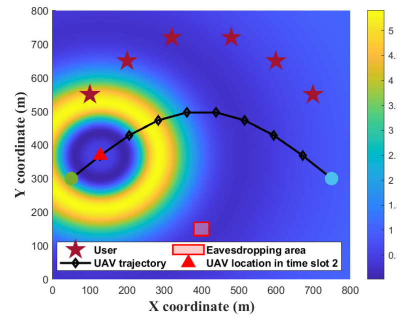

Fig. 4: Beamforming pattern gain achieved under the FPA scheme.

Therefore, constraints (9), (12), (14), and (15) can be reformulated as

$$\varpi_m[n] \geq \sum_{i\neq m}^{\mathcal{M}} \beta_m[n]\phi_{m,i}^{\text{ub}}[n] + \sigma_m^2, \tag{33}$$

$$\sum_{i=1}^{\mathcal{M}} \beta_m[n]\phi_{m,i}^{\text{lb}}[n] + \sigma_m^2 \geq \frac{1}{2}\left(\frac{\omega_m^{(l)}[n]}{\varpi_m^{(l)}[n]}\varpi_m^2[n] + \frac{\varpi_m^{(l)}[n]}{\omega_m^{(l)}[n]}\omega_m^2[n]\right), \tag{34}$$

$$\sum_{i\neq m}^{\mathcal{M}} \beta_j[n]\phi_{j,i}^{\text{lb}}[n] + \sigma_j^2 \geq \frac{1}{2}\left(\frac{\tilde{\omega}_{m,j}^{(l)}[n]}{\tilde{\varpi}_j^{(l)}[n]}\tilde{\varpi}_j^2[n] + \frac{\tilde{\varpi}_j^{(l)}[n]}{\tilde{\omega}_{m,j}^{(l)}[n]}\tilde{\omega}_{m,j}^2[n]\right), \tag{35}$$

$$\tilde{\varpi}_j[n] \geq \sum_{i=1}^{\mathcal{M}} \beta_j[n]\phi_{j,i}^{\text{ub}}[n] + \sigma_j^2. \tag{36}$$

With the above transformations, problem (**P4**) can be reformulated as the following convex optimization problem

$$(\textbf{P4-1}): \max_{x_k[n],\Phi} \sum_{m=1}^{M}\left(\sum_{n=1}^{N}\tilde{R}_m^{\text{lb}}[n] + \eta_m\right) \tag{37a}$$

$$\text{s.t. } (6c)-(6d), (16), (24), (31), (32)-(36).$$

IV. Numerical Results

This section presents numerical results to demonstrate the effectiveness of FAs in the secure UAV communication system. An $800\ \text{m}\times 800\ \text{m}$ square area with $M=6$ ground users is considered, while the eavesdropper is assumed to lie within a 40 m × 40 m region, in which $J=9$ candidate locations are sampled. The UAV flies at a fixed altitude of $H=100$ m over a mission period of $T=25$ s, which is divided into $N=10$ equal time slots of duration $\tau=2.5$ s. The UAV is equipped with $K=6$ antennas, and the maximum transmit power is limited to $P_{\max}=30$ dBm. The channel power gain at the reference distance of 1 m is set to $h_0=-50$ dB, while the noise power is -100 dBm. For the FA configuration, the antenna mobility region is $[0, 20\lambda]$ with a minimum inter-antenna spacing of 0.5λ, where the carrier wavelength is $\lambda=0.0107$ m. The FA movement duration within each slot is $t_1[n]=0.2$ s with a movement speed of 0.1 m/s.

Fig. 2 illustrates the secrecy sum-rate performance of the proposed scheme and the FPA scheme versus the maximum transmit power for $K=6$ and $K=8$. It can be observed that the secrecy sum rate of all schemes increases monotonically with the transmit power, since the received signal power at both the users and the potential eavesdroppers is simultaneously enhanced. Meanwhile, the proposed scheme exhibits a consistently steeper growth trend over the entire power range, indicating a more efficient utilization. For both antenna configurations, the proposed scheme achieves a noticeably enhanced secrecy sum-rate performance relative to the corresponding FPA scheme. Moreover, increasing the number of FAs from 6 to 8 provides an additional gain of about 20 bits/Hz at high transmit power, demonstrating favorable scalability.

Fig. 3 and Fig. 4 compare the spatial beamforming gain distributions of the proposed scheme and the FPA scheme, where the predefined UAV trajectory and the locations of users and the potential eavesdropping region are indicated. It can be observed that, under the FA scheme, the radiated energy is adaptively concentrated toward the users, while pronounced low-gain regions are formed over the eavesdropping area, exhibiting a highly directive and reconfigurable beam pattern. This results from the joint optimization of the transmit beamforming and the FA positions, which enables the spatial radiation pattern to dynamically adapt to the UAV location. By contrast, the FPA scheme produces a nearly isotropic beamforming gain distribution. Although basic coverage for users can be maintained, effective spatial suppression toward the eavesdropping region cannot be achieved, leading to signal energy leakage toward unintended directions.

V. Conclusion

In this paper, we have investigated the FA-enabled UAV communication system to enhance the PLS under uncertain eavesdropper locations. To improve the secrecy performance, the UAV equipped with an FA array serves multiple users, where the secrecy sum-rate is maximized through the joint optimization of the transmit beamforming and the FA positions. To deal with the non-convexity of the formulated optimization problem, an AO-based algorithm is proposed to iteratively update the variables. Simulation results demonstrate that the proposed FA-enabled design significantly improves the secrecy performance.

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